

Aayush Singh

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GitHub — LinkedIn

Professional Summary

Information Technology student pursuing Honors in Artificial Intelligence and Machine Learning with hands-on experience in Python, SQL, pandas, data preprocessing, ETL pipelines, and cloud-based data engineering. Built end-to-end data engineering and machine learning projects involving data ingestion, transformation, storage, analysis, and visualization. Strong foundation in data processing and API-based workflows with an interest in building reliable data pipelines and AI-integrated systems.

Education

Shah & Anchor Kutchhi Engineering College, Mumbai

Bachelor of Engineering (Information Technology)

CGPA: 8.5+

Expected Graduation: 2027

Technical Skills

Languages: Python, SQL, Java, C++.

Data Engineering: ETL, Data Ingestion, Data Transformation, Data Cleaning, Data Processing

Data & ML: pandas, NumPy, scikit-learn, Matplotlib, LightGBM, Random Forest, Linear Regression

AWS: S3, Glue, Athena, Lambda, API Gateway, DynamoDB, SNS

Databases & Tools: MySQL, MongoDB, Git, GitHub, Postman

Projects

YouTube Data Analysis

End-to-End Data Engineering

- Built an end-to-end ETL pipeline for processing and analyzing YouTube trending datasets.
- Used Python and AWS Glue to ingest and transform raw data into structured datasets.
- Stored raw and processed datasets in Amazon S3 as a cloud-based data lake.
- Used Amazon Athena to query processed datasets for analytical insights.
- Created dashboards in Amazon QuickSight to visualize trends and key metrics.

Twitter/X Data Pipeline

Data Engineering Project

- Built a data pipeline to collect, process, and analyze Twitter/X data.
- Implemented data processing steps to transform raw data into structured datasets.
- Applied data cleaning and transformation techniques to prepare data for analysis.

Energy Demand Forecasting

Python, pandas, scikit-learn

- Processed and analyzed Delhi energy-demand data using Python and pandas.
- Performed data preprocessing and exploratory data analysis before model development.
- Built Random Forest and Linear Regression models for energy demand prediction.
- Evaluated model performance using regression evaluation metrics.

AssignHub

AWS Serverless Platform

- Built a serverless application using AWS Lambda, API Gateway, DynamoDB, S3, and SNS.
- Processed multipart/form-data submissions through APIs and stored files in Amazon S3.
- Stored structured submission metadata in DynamoDB for persistent data management.
- Automated email notifications using Amazon SNS as part of the application workflow.

Achievements

- Achieved **Global Rank 2680** in the Amazon ML Challenge 2025 for product price prediction.

Certifications

Palo Alto Networks — Cybersecurity Fundamentals — Network Security Fundamentals — Fundamentals of Cloud Security